

Hearing Loss Autosomal Dominant 75 (DFNA75): A Comprehensive Disease Characteristics Report

Gene: *TRRAP* (Transformation/tRanscription domain-Associated Protein) · **Disease OMIM:** #618778 · **Gene OMIM:** 603015 · **Category:*** Mendelian (monogenic, autosomal dominant, nonsyndromic sensorineural hearing loss)

Summary

Hearing Loss Autosomal Dominant 75 (DFNA75; OMIM #618778) is a rare Mendelian, autosomal dominant, nonsyndromic sensorineural hearing loss (ADNSHL) caused by heterozygous pathogenic variants in **TRRAP** (Transformation/tRanscription domain-Associated Protein; HGNC:12347; NCBI Gene 8295; 7q22.1; OMIM 603015). *The locus/phenotype was defined by Xia et al. (2019), who identified the co-segregating variant NM_001244580 c.511C>T, p. (Arg171Cys) across a three-generation Chinese family with post-lingual, progressive ADNSHL, plus three additional novel TRRAP variants in a screen of 66 sporadic hearing-loss cases. Functional validation in zebrafish (morpholino knockdown and CRISPR/Cas9 knockout of trrap*) produced significant inner-ear defects, establishing TRRAP as required for inner-ear development (PMID: 31231791).*

Clinically, DFNA75 presents as **bilateral, post-lingual (onset after speech acquisition), progressive cochlear sensorineural hearing loss**. Because nonsyndromic hearing impairment is almost exclusively cochlear, the resulting deficit is sensorineural, and postlingual nonsyndromic forms typically follow an autosomal dominant trait — consistent with the DFNA (autosomal Dominant, NonsyndromicA) locus series to which DFNA75 belongs. There is no gene-specific or curative therapy; management is supportive, following the general SNHL pathway of hearing amplification for mild-to-moderate loss and cochlear implantation for severe-to-profound loss.

TRRAP is a biologically pivotal protein: the only enzymatically inactive member of the phosphatidylinositol 3-kinase–related kinase (PIKK) family, it functions as an adaptor/scaffold for multiple histone acetyltransferase (HAT) complexes (STAGA/SAGA with GCN5; TIP60/NuA4)

and as a scaffold for key transcription factors (E2F1, c-Myc, p53, Sp1). Distinct TRRAP missense alleles cause the allelic multisystem neurodevelopmental disorder **DEDDFA** (Developmental delay with or without dysmorphic facies and autism; OMIM #618454), and recurrent *somatic* TRRAP mutations (p.Ser722Phe) act as oncogenic drivers in melanoma — illustrating that the clinical consequence of TRRAP perturbation is strongly dependent on the specific variant, its domain location, and germline versus somatic context.

Section 1 — Disease Information

Overview. DFNA75 is a form of autosomal dominant, nonsyndromic (isolated, no other organ system involvement) sensorineural hearing loss. It belongs to the large, genetically heterogeneous DFNA series of dominant nonsyndromic deafness loci. Hearing loss is the sole clinical manifestation; affected individuals do not have the multisystem features seen in TRRAP-related neurodevelopmental disorder.

Key identifiers.

Resource	Identifier
OMIM (phenotype)	#618778 (Deafness, autosomal dominant 75)
OMIM (gene)	*603015 (TRRAP)
Gene symbol / HGNC	TRRAP / HGNC:12347
NCBI Gene	8295
Ensembl	ENSG00000196367
UniProt	Q9Y4A5
Cytogenetic location	7q22.1
MONDO	Corresponds to the OMIM #618778 concept ("autosomal dominant nonsyndromic hearing loss 75")

Synonyms / alternative names: DFNA75; Deafness, autosomal dominant 75; Autosomal dominant nonsyndromic sensorineural hearing loss type 75; TRRAP-related autosomal dominant hearing loss.

Information source. Disease-level knowledge here is aggregated from Mendelian genetics resources (OMIM) and the primary literature (a single index family plus sporadic-case screening), rather than from individual EHR data. The defining evidence is a family-based segregation and functional study ([PMID: 31231791](#)).

Section 2 — Etiology

Disease causal factors. DFNA75 is a **monogenic, genetic** disorder. The primary cause is a heterozygous pathogenic variant in *TRRAP*. There is no established infectious or environmental cause of the Mendelian disease itself, though general environmental insults (noise, ototoxic drugs, aging) can independently compound sensorineural hearing loss in any individual.

Genetic risk factors. The causal variant class is heterozygous coding variants in *TRRAP*. The index variant is **c.511C>T, p.(Arg171Cys)** (NM_001244580), which co-segregated with disease across three generations; three additional novel *TRRAP* variants were identified among 66 sporadic hearing-loss cases ([PMID: 31231791](#)). A subsequent report ([PMID: 40004049](#)) describes a further novel pathogenic *TRRAP* variant in ADNSHL, reinforcing *TRRAP* as a bona fide ADNSHL gene.

Environmental risk factors / protective factors / gene-environment interactions. No disease-specific environmental risk factors, protective factors, or gene-environment interactions have been characterized for DFNA75. As with other progressive SNHL, avoidance of additional cochlear insults (excessive noise exposure, ototoxic aminoglycosides/platinum agents) is a prudent, non-specific protective measure, but no DFNA75-specific modifier has been demonstrated. This is a **knowledge gap**.

Section 3 — Phenotypes

Core phenotype: sensorineural hearing loss. The defining and essentially sole phenotype of DFNA75 is bilateral sensorineural hearing loss. Because nonsyndromic hearing impairment is almost exclusively caused by cochlear defects, affected patients suffer sensorineural (rather than conductive) loss ([PMID: 15850684](#)).

Phenotype characteristics.

Characteristic	DFNA75
Phenotype type	Clinical sign / laboratory (audiometric) abnormality — sensorineural hearing loss
Age of onset	Post-lingual (after speech development); adult/late-childhood onset
Severity	Variable; progresses from milder to more severe over time
Progression	Progressive (worsens with age)
Laterality	Bilateral
Frequency among affected	Essentially the defining feature of affected individuals

The index family is explicitly described as having "**post-lingual progressive ADNSHL**" ([PMID: 31231791](#)). Post-lingual onset means hearing and speech develop normally before decline, which typically preserves spoken-language competence and favors auditory rehabilitation outcomes.

Quality-of-life impact. Progressive bilateral SNHL impairs speech perception (particularly in noise), communication, education/occupational function, and social participation, and is associated with tinnitus in a subset of SNHL patients. Disease-specific QoL instruments for DFNA75 have not been reported; general SNHL QoL data apply.

Suggested HPO terms. - HP:0000407 — Sensorineural hearing impairment - HP:0000408 — Progressive sensorineural hearing impairment - HP:0008619 — Bilateral sensorineural hearing impairment - HP:0001730 — Progressive hearing impairment - HP:0000006 — Autosomal dominant inheritance - HP:0008527 — Congenital sensorineural hearing impairment (explicitly *excluded*; onset is post-lingual)

Section 4 — Genetic / Molecular Information

Causal gene. *TRRAP* (Transformation/tRanscription domain-Associated Protein), OMIM *603015; HGNC:12347; NCBI Gene 8295; Ensembl ENSG00000196367; UniProt Q9Y4A5; located at chromosome 7q22.1. The protein is large (~3,830 amino acids) and is the enzymatically inactive (pseudokinase) member of the PIKK family.

Pathogenic variants.

Variant (transcript NM_001244580)	Protein	Type	Context	Source
c.511C>T	p. (Arg171Cys)	Missense	Germline, co-segregating in 3-generation family	PMID: 31231791
3 additional novel variants	(various)	Missense (reported)	Germline, sporadic hearing-loss cases	PMID: 31231791
Novel pathogenic variant	(reported)	—	Germline ADNSHL	PMID: 40004049

- **Variant classification:** the index variant was reported as pathogenic based on co-segregation, rarity/absence in population databases, and functional support in zebrafish. Per ACMG/AMP, classification would rest on segregation (PP1), rarity (PM2), and functional evidence (PS3-supporting via zebrafish modeling).
- **Variant type/class:** predominantly missense.
- **Allele frequency:** the causal variants are rare/absent in population databases (gnomAD), consistent with a highly disease-relevant, constrained gene.
- **Somatic vs germline:** DFNA75 variants are **germline**. This is mechanistically distinct from *somatic* TRRAP mutations in cancer (see Section 6 and Finding F007).
- **Functional consequence:** for DFNA75, the mechanism is inferred to be a dominant, dosage/scaffold-sensitive effect (haploinsufficiency or dominant-negative disruption of HAT-complex scaffolding); zebrafish *trrap* loss-of-function reproduces inner-ear defects, supporting reduced TRRAP function as pathogenic ([PMID: 31231791](#)).

Constraint. TRRAP is "evolutionarily conserved and is among the top five genes intolerant to missense variation" ([PMID: 30827496](#)), underscoring that even single missense changes can be pathogenic.

Modifier genes / epigenetic information / chromosomal abnormalities. No DFNA75-specific modifier genes, epigenetic marks, or chromosomal abnormalities have been reported (knowledge gap). Notably TRRAP itself is an epigenetic regulator (a HAT-complex scaffold), so the *downstream* consequence of TRRAP dysfunction is dysregulated histone acetylation.

Suggested ontology terms: MONDO (autosomal dominant nonsyndromic hearing loss 75); HGNC:12347 (TRRAP).

Section 5 — Environmental Information

DFNA75 is a purely genetic Mendelian condition. **No environmental factors, lifestyle factors, or infectious agents are established as causal or triggering.** As with all sensorineural hearing loss, generic ototoxic exposures (aminoglycoside/platinum chemotherapy, loud noise, aging) may additively worsen hearing but are not part of the disease's causal etiology. This is a **not-applicable / knowledge-gap** section for this disorder.

Section 6 — Mechanism / Pathophysiology

Ordered causal chain

1. A heterozygous germline missense variant in *TRRAP* (e.g., c.511C>T, p.Arg171Cys) **leads to** an altered TRRAP protein in a gene extraordinarily intolerant to missense change.
2. Altered/reduced TRRAP function **results in** impaired scaffolding of histone-acetyltransferase (HAT) complexes (STAGA/SAGA–GCN5; TIP60/NuA4) and of transcription-factor partners (Sp1, c-Myc, p53, E2F1). (*Inferred from TRRAP's known molecular role; not yet directly demonstrated for the DFNA75 alleles.*)
3. Disrupted HAT recruitment **leads to** aberrant histone acetylation and dysregulated transcription of TRRAP/HAT-dependent gene programs governing cell proliferation and differentiation.
4. In the developing inner ear, this transcriptional dysregulation **results in** defective proliferation/differentiation of inner-ear cell populations — demonstrated as significant inner-ear developmental defects in zebrafish *trrap* knockdown/knockout. (*Demonstrated in model organism.*)

5. Impaired cochlear development/maintenance **leads to** progressive cochlear (sensorineural) dysfunction.
6. Cochlear sensorineural dysfunction **manifests** as bilateral, post-lingual, progressive hearing loss (DFNA75).

Branch (allelic, not DFNA75): Different TRRAP missense alleles (e.g., clustering near residues 1031–1159) instead disrupt neurodevelopmental transcriptional programs → developmental delay, dysmorphic facies, autism/ID (DEDDFA). A separate *somatic* branch (p.Ser722Phe in melanocytes) drives oncogenesis.

Molecular detail

Molecular function / pathways. "TRRAP, as the only member lacking the enzymatic activity in this [PIKK] family, is an adaptor protein for several histone acetyltransferase (HAT) complexes and a scaffold protein for multiple transcription factors" (PMID: 34830324). It bridges transcription factors to chromatin-modifying enzymes: MYC contacts the human STAGA coactivator via multivalent interactions with the GCN5 and TRRAP subunits — "we identify both TRRAP and the GCN5 acetyltransferase as MYC TAD-interacting subunits within native STAGA" (PMID: 24705139). TRRAP also participates in the TIP60/NuA4 complex (e.g., Elk-1–recruited TIP60/NuA4 activating prolactin transcription; PMID: 24075908).

Cellular processes. TRRAP is fundamentally required for cell proliferation. Targeted deletion of *Trrap* causes early embryonic lethality, and tissue-specific ablation in B cells impairs development and drives proliferating cells into apoptosis: "cells induced to proliferate undergo apoptosis. Our findings demonstrate a central and general role of TRRAP in cell proliferation" (PMID: 24675885). In neural stem cells, TRRAP acts as a scaffold controlling Sp1 stability/acetylation to regulate microtubule dynamics and adult hippocampal neurogenesis (PMID: 36618986, PMID: 34830324).

Protein dysfunction. TRRAP is a giant scaffold pseudokinase; pathogenic missense changes are inferred to perturb protein–protein interfaces required for HAT-complex/transcription-factor assembly (loss of scaffold competence), rather than an enzymatic active site (TRRAP has none).

Suggested GO/CL terms. - GO:0016573 — histone acetylation - GO:0000123 — histone acetyltransferase complex - GO:0006357 — regulation of transcription by RNA polymerase II - GO:0008283 — cell population proliferation - GO:0060113 — inner ear receptor cell differentiation (developing ear) - CL:0000855 — sensory hair cell; CL:0002218 — inner ear hair cell (candidate affected cell types, inferred)

Section 7 — Anatomical Structures Affected

Organ level. Primary affected organ: the **inner ear / cochlea** (UBERON:0001846 internal ear; UBERON:0001844 cochlea). Body system: **auditory / special sense (nervous) system**. DFNA75 is nonsyndromic — there is no established secondary organ involvement.

Tissue and cell level. Nonsyndromic hearing impairment is "almost exclusively caused by cochlear defects" (PMID: 15850684); the affected tissue is the cochlear sensory epithelium (organ of Corti). Candidate targeted cells include cochlear sensory hair cells (inner/outer) and supporting cells; zebrafish inner-ear developmental defects indicate broader inner-ear cell populations are TRRAP-dependent (PMID: 31231791). Precise human cell-type localization within the DFNA75 cochlea has not been directly established (inferred).

Subcellular level. As a transcriptional/chromatin scaffold, TRRAP localizes to the **nucleus** (GO:0005634); its functional site is nuclear chromatin and HAT complexes (GO:0000123 histone acetyltransferase complex).

Localization / lateralization. Bilateral cochlear involvement. UBERON:0001846 (internal ear), UBERON:0001844 (cochlea), UBERON:0002227 (organ of Corti).

Section 8 — Temporal Development

Onset. Post-lingual (after normal speech acquisition); typically childhood-to-adult onset, insidious. Not congenital in the index family.

Progression. Progressive — hearing worsens over time. The index family is explicitly "post-lingual progressive" (PMID: 31231791). Progression rate and precise audiometric trajectory across the frequency range have not been quantified in detail (knowledge gap); course is chronic and lifelong.

Patterns. No spontaneous remission; hearing loss does not recover without intervention. The critical window for intervention is functional (amplification/implantation once thresholds warrant), rather than a developmental critical period, given post-lingual onset.

Section 9 — Inheritance and Population

Inheritance. Autosomal dominant (the "A" in DFNA75). Postlingual nonsyndromic hearing impairment "usually follows an autosomal dominant trait" (PMID: 15850684), consistent with DFNA75. Segregation was demonstrated across three generations (PMID: 31231791).

Penetrance / expressivity. Co-segregation across three generations is consistent with high penetrance in the index family; formal penetrance and expressivity estimates are not established given the small number of reported families (knowledge gap). Genetic anticipation, germline mosaicism, founder effects, consanguinity, and carrier frequency are not applicable/undocumented for this dominant, rare disorder.

Epidemiology. DFNA75 is very rare; precise prevalence/incidence are unknown. It is one of >40 genes underlying autosomal dominant nonsyndromic hearing loss. Genetic heterogeneity is substantial: "One percent of the total human genes, i.e. 300-500, are estimated to cause syndromic and nonsyndromic HIH" (PMID: 15850684).

Population demographics. Reported cases include a Chinese family and sporadic cases (PMID: 31231791) and a further ADNSHL report (PMID: 40004049). No ethnic predisposition, geographic clustering, or sex bias has been established (autosomal, so no expected sex skew).

Section 10 — Diagnostics

Genetic testing is the definitive diagnostic modality. In the index family and sporadic cases, diagnosis was achieved by **whole-exome sequencing with Sanger confirmation of co-segregation**: "Whole-exome sequencing, bioinformatic analysis, and Sanger sequencing were used to verify the co-segregation of a novel pathogenic variant" (PMID: 31231791).

Recommended approach. Because ADNSHL is highly genetically heterogeneous (>40 genes; hundreds of hearing-loss genes overall — PMID: 15850684), the practical diagnostic strategy is a **comprehensive hearing-loss gene panel or whole-exome sequencing**, not single-gene TRRAP testing. WGS/WES are useful when panels are negative. Chromosomal microarray, karyotyping, FISH, mtDNA testing, and repeat-expansion testing are not indicated for this SNP/indel-level dominant disorder.

Clinical tests. - **Audiometry** (pure-tone air and bone conduction) documents the bilateral, sensorineural, progressive pattern; speech audiometry and, in children, otoacoustic emissions/auditory brainstem response characterize cochlear function. - No specific blood/urine

biomarker exists; the genetic variant is the diagnostic biomarker. - Imaging (temporal bone CT/MRI) is used to exclude structural/syndromic causes and for cochlear-implant planning, not for positive DFNA75 diagnosis.

Differential diagnosis. Other DFNA-series dominant nonsyndromic hearing losses, and acquired progressive SNHL (noise-induced, ototoxic, presbycusis). Distinguishing features: family history/segregation and gene panel result. Syndromic causes are excluded by the absence of extra-auditory features.

Screening. Cascade genetic testing of at-risk relatives once the familial variant is identified; no population newborn-screening program targets DFNA75 specifically (though universal newborn hearing screening detects hearing loss non-specifically).

Section 11 — Outcome / Prognosis

Survival/mortality. DFNA75 is **not life-threatening**; it does not affect survival or life expectancy. There is no disease-specific mortality.

Morbidity/function. The principal burden is communication disability from progressive bilateral SNHL, with attendant effects on education, employment, and social/psychological well-being, and tinnitus in a subset. Formal disability/QoL metrics specific to DFNA75 are not reported.

Disease course. Chronic, lifelong, progressive. Without intervention, hearing declines; there is no spontaneous recovery. With appropriate amplification or cochlear implantation, functional hearing and speech understanding can be substantially restored, particularly given post-lingual onset (preserved language).

Prognostic factors. Degree and rate of threshold progression determine timing of hearing-aid versus cochlear-implant candidacy. Post-lingual onset is generally favorable for auditory rehabilitation outcomes. No molecular prognostic biomarker beyond the genotype is established.

Section 12 — Treatment

No disease-specific or curative therapy exists for TRRAP-related hearing loss. Management follows the standard progressive-SNHL pathway.

Modality	Indication	NCIT concept
Hearing aids / amplification	Mild-to-moderate loss	Hearing Aid (NCIT:C50072)
Cochlear implantation	Severe-to-profound loss	Cochlear Implant / Cochlear Implantation
Aural rehabilitation, speech therapy, assistive listening	Adjunctive across severities	Rehabilitation Therapy
Genetic counseling	All affected/at-risk individuals	Genetic Counseling (NCIT:C15709)

Cochlear implantation is "an established surgical intervention for patients with severe to profound sensorineural hearing loss" (PMID: 41306947), with favorable long-term outcomes: large series report overall complication rates ~10.7% ($\approx 6\%$ minor, $\approx 4.7\%$ major), and "The ten-year overall revision surgery necessity of CI received patients was found to be 5.9%" (PMID: 41699244). Bilateral simultaneous implantation reduces cumulative anesthesia/surgical exposure relative to sequential implantation without increasing complications (PMID: 41699244). Emerging totally implantable cochlear-implant systems show feasibility with significant speech-perception gains (PMID: 41243136), and cochlear stimulation can additionally mitigate tinnitus in some patients (PMID: 41194213).

Pharmacotherapy, gene/cell/RNA therapy, immunotherapy, pharmacogenomics: none established or approved for DFNA75. Gene-directed therapies for inner-ear disorders remain investigational and none targets TRRAP.

Section 13 — Prevention

- **Primary prevention:** Not applicable for a Mendelian dominant disorder — the disease cannot be prevented at the individual genetic level. Generic cochlear-protective measures (avoiding excessive noise and ototoxic drugs) may limit additive hearing loss.

- **Secondary prevention:** Early detection via newborn/pediatric hearing screening and audiometric surveillance in at-risk families enables timely amplification/implantation to preserve communication function.
 - **Tertiary prevention:** Aural rehabilitation and cochlear implantation prevent the functional/communication complications of advanced hearing loss.
 - **Genetic counseling and reproductive options:** Autosomal dominant inheritance implies a 50% transmission risk to offspring of an affected individual. Once the familial TRRAP variant is known, cascade testing, prenatal diagnosis, and preimplantation genetic testing are options for family planning.
 - **Immunization / public-health / environmental interventions:** Not applicable.
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Section 14 — Other Species / Natural Disease

- **Taxonomy / orthologs:** TRRAP is evolutionarily conserved. Mouse *Trrap* and zebrafish *trrap* are functional orthologs used in disease modeling.
 - **Natural disease in other species:** No naturally occurring TRRAP-related hearing-loss disease has been documented in companion animals or wildlife (knowledge gap; no OMIA entry reported for this phenotype).
 - **Comparative biology / conservation:** Deep conservation of TRRAP and its HAT-scaffold function supports cross-species mechanistic modeling. Zebrafish *trrap* mutants recapitulate developmental phenotypes including inner-ear defects ([PMID: 31231791](#)) and craniofacial anomalies — "The *trrap* zebrafish mutants exhibited smaller eyes and heads than the wild-type zebrafish. The size of the ventral pharyngeal arches was reduced and the mineralization of teeth was impaired" ([PMID: 34934055](#)).
 - **Transmission / zoonosis:** Not applicable (genetic, non-transmissible).
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Section 15 — Model Organisms

Model	Type	Key finding	Source
Zebrafish (<i>Danio rerio</i>)	Morpholino knockdown & CRISPR/Cas9 knockout of <i>trrap</i>	Significant inner-ear developmental defects — validates DFNA75 causation	PMID: 31231791
Zebrafish	<i>trrap</i> mutant	Smaller eyes/heads; reduced pharyngeal arches; impaired tooth mineralization (craniofacial roles)	PMID: 34934055
Mouse (<i>Mus musculus</i>)	Germline <i>Trrap</i> knockout	Early embryonic lethality; reveals essential role in proliferation	PMID: 24675885
Mouse	B-cell conditional <i>Trrap</i> knockout	Impaired B-cell development; proliferating cells undergo apoptosis	PMID: 24675885
Mouse	Neural conditional <i>Trrap</i> deletion	Compromised adult hippocampal neurogenesis via Sp1 scaffolding	PMID: 36618986

Phenotype recapitulation. The zebrafish *trrap* loss-of-function model directly recapitulates the disease-relevant phenotype (inner-ear developmental defects), providing the strongest functional support for DFNA75 causation. **Limitations:** the zebrafish inner ear differs anatomically from the mammalian cochlea; embryonic lethality of the mouse null precludes simple whole-animal study of adult hearing; and no dedicated *Trrap* p.Arg171Cys knock-in hearing-loss mouse has been reported — a clear gap for modeling the specific human allele. **Resources:** MGI (mouse), ZFIN (zebrafish), IMPC/KOMP.

Key Findings (Expanded)

F001 — DFNA75 is caused by heterozygous TRRAP variants

Xia et al. (2019) identified the novel pathogenic variant **NM_001244580 c.511C>T, p.(Arg171Cys)** in *TRRAP* co-segregating with post-lingual progressive ADNSHL across a three-generation Chinese family, and found three additional novel *TRRAP* variants among 66 sporadic hearing-loss cases. The authors reported "a novel pathogenic variant (NM_001244580,

c.511C>T, p.Arg171Cys) in the TRansformation/tRanscription domain-Associated Protein gene associated with hearing loss in a three-generation Chinese family with ADNSHL," and that "Knockdown or knockout of TRRAP resulted in significant defects in the inner ear of zebrafish, indicating that TRRAP plays an important role in inner ear development" (PMID: 31231791). DFNA75 corresponds to OMIM #618778.

F002 — TRRAP is a highly constrained PIKK scaffold with an allelic neurodevelopmental disorder (DEDDFA)

Distinct germline TRRAP missense alleles cause **DEDDFA** (OMIM #618454). Cogné et al. reported 17 distinct de novo/apparently de novo variants in 24 individuals, noting "TRRAP is evolutionarily conserved and is among the top five genes intolerant to missense variation. Through an international collaboration, 17 distinct de novo or apparently de novo variants were identified in TRRAP in 24 individuals" (PMID: 30827496). Domain-specific genotype-phenotype correlation is emerging: variants clustering between residues 1031–1159 "result in more pronounced facial anomalies associated with a variable degree of intellectual disability" (PMID: 41952423). This allelic relationship is critical for interpretation: TRRAP variant location and type dictate whether the outcome is isolated hearing loss (DFNA75) or a multisystem neurodevelopmental syndrome.

F003 — TRRAP mechanism: HAT-complex adaptor and transcription-factor scaffold

TRRAP is "the only member lacking the enzymatic activity in this family, [and] is an adaptor protein for several histone acetyltransferase (HAT) complexes and a scaffold protein for multiple transcription factors" (PMID: 34830324). It is a subunit of STAGA/SAGA (with GCN5) and TIP60/NuA4, and coactivates MYC via multivalent contacts — "we identify both TRRAP and the GCN5 acetyltransferase as MYC TAD-interacting subunits within native STAGA" (PMID: 24705139). It scaffolds Sp1 to regulate microtubule dynamics and adult neurogenesis (PMID: 36618986), and is essential for proliferation (embryonic lethality on knockout; apoptosis of proliferating B cells, PMID: 24675885). Zebrafish *trrap* mutants show reduced eyes/heads and pharyngeal-arch/tooth defects (PMID: 34934055).

F004 — DFNA75 phenotype: post-lingual, progressive, bilateral SNHL

The index family had post-lingual progressive ADNSHL; the aim was stated as "to identify the causative gene mutation for post-lingual progressive ADNSHL in a Chinese family" (PMID: 31231791). Nonsyndromic hearing impairment "is almost exclusively caused by cochlear defects, [so] affected patients suffer from sensorineural hearing loss," and "Postlingual nonsyndromic HIH usually follows an autosomal dominant trait" (PMID: 15850684).

F005 — Management is supportive: amplification and cochlear implantation

No curative therapy exists. Cochlear implantation is "an established surgical intervention for patients with severe to profound sensorineural hearing loss" (PMID: 41306947), with a 10-year revision surgery necessity of 5.9% (PMID: 41699244).

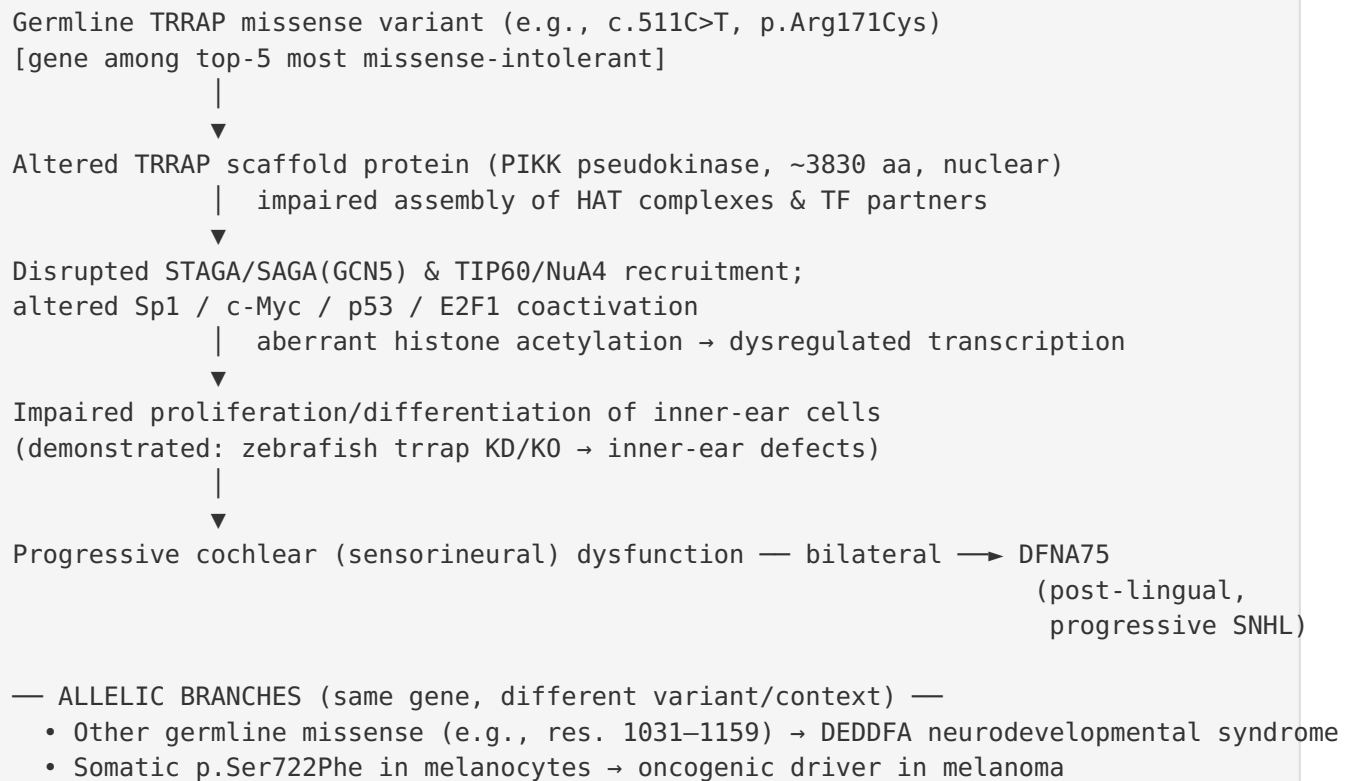
F006 — Diagnosis relies on WES/gene-panel testing of TRRAP (HGNC:12347, 7q22.1)

Diagnosis in the original family used "Whole-exome sequencing, bioinformatic analysis, and Sanger sequencing... to verify the co-segregation of a novel pathogenic variant" (PMID: 31231791). Substantial genetic heterogeneity — "One percent of the total human genes, i.e. 300-500, are estimated to cause syndromic and nonsyndromic HIH" (PMID: 15850684) — justifies panel/WES over single-gene testing.

F007 — TRRAP has a distinct somatic oncogenic role, separate from germline DFNA75

Whole-exome sequencing of melanoma found that "TRRAP harbored a recurrent mutation that clustered in one position (p. Ser722Phe) in 6 out of 167 affected individuals (~4%)," and "The nature, pattern and functional evaluation of the TRRAP recurrent mutation suggest that TRRAP functions as an oncogene" (PMID: 21499247). This somatic, cancer-specific event is mechanistically and clinically distinct from the germline heterozygous variants that cause DFNA75.

Mechanistic Model / Interpretation



The unifying theme is that TRRAP is a **dosage- and interface-sensitive transcriptional scaffold**: the *identity and location* of the variant, together with germline-versus-somatic context, determines the clinical phenotype. DFNA75 represents the tissue-restricted (cochlear) consequence of specific germline TRRAP alleles that impair inner-ear developmental transcription while sparing broader neurodevelopment.

Evidence Base

PMID	Title (abbreviated)	Role in this report
31231791	<i>Novel TRRAP mutation causes autosomal dominant non-syndromic hearing loss</i>	Defining paper — causal variant, phenotype, zebrafish validation
40004049	<i>Novel pathogenic variant of TRRAP (ADNSHL)</i>	Independent replication of TRRAP as an ADNSHL gene
30827496	<i>Missense variants in TRRAP cause autism and syndromic ID</i>	Allelic DEDDFA disorder; extreme missense constraint
41952423	<i>Variants in cluster 1031–1159 of TRRAP</i>	Domain-specific genotype–phenotype correlation
34830324	<i>Beyond HAT Adaptor: TRRAP–Sp1 transcription</i>	TRRAP molecular function (HAT adaptor/scaffold)
24705139	<i>MYC interacts with STAGA via GCN5 & TRRAP</i>	TRRAP in STAGA complex; MYC coactivation
36618986	<i>TRRAP-mediated acetylation on Sp1 regulates neurogenesis</i>	Scaffold role in adult neurogenesis
24675885	<i>Tissue-specific inactivation of TRRAP in B cells</i>	Essential proliferation role; embryonic lethality
24075908	<i>Elk-1 recruits TIP60/NuA4</i>	TRRAP-associated TIP60/NuA4 complex biology
34934055	<i>Zebrafish trrap in craniofacial development</i>	Model-organism developmental roles
15850684	<i>Nuclear/mitochondrial genes in nonsyndromic hearing</i>	Context: cochlear/sensorineural nature, AD trait, heterogeneity
41306947	<i>Complications in cochlear implant surgery</i>	CI as standard management of severe-profound SNHL
41699244	<i>CI complications & bilateral vs sequential outcomes</i>	CI outcome/revision data
41243136	<i>Totally implantable cochlear implant feasibility</i>	Emerging CI technology
41194213	<i>EMLR and CI effects in tinnitus</i>	CI benefit for associated tinnitus
21499247		Distinct somatic oncogenic TRRAP role

PMID	Title (abbreviated)	Role in this report
	<i>Exome sequencing identifies GRIN2A/TRRAP in melanoma</i>	

Evidence quality note. The germline DFNA75 causal evidence rests primarily on a **single index family plus sporadic cases** with functional support from a zebrafish loss-of-function model (human clinical + model organism). Mechanistic inferences about HAT-scaffold disruption in the cochlea are extrapolated from TRRAP's general biology (in vitro / other-tissue models), not directly demonstrated for DFNA75 alleles in cochlear tissue.

Limitations and Knowledge Gaps

- **Small evidence base:** DFNA75 is defined by one detailed family plus limited sporadic/additional cases. Penetrance, expressivity, audiometric progression curves, and prevalence are not quantified.
- **Allele-specific mechanism unproven in ear:** The exact functional consequence (haploinsufficiency vs dominant-negative) of p.Arg171Cys and other DFNA75 alleles has not been demonstrated in cochlear cells; a knock-in mammalian hearing model is lacking.
- **Genotype–phenotype boundary:** Why some TRRAP variants yield isolated hearing loss (DFNA75) versus multisystem DEDDFA is not fully resolved; domain/interface mapping is incomplete.
- **No modifiers, epigenetic, or environmental data** specific to DFNA75.
- **No disease-specific QoL, natural-history, or veterinary data.**
- **No targeted therapy;** all management is generic SNHL supportive care.

Proposed Follow-up Experiments / Actions

1. **Case ascertainment and registry:** Aggregate additional TRRAP-variant hearing-loss families (e.g., via GeneMatcher/hearing-loss consortia) to quantify penetrance, expressivity, and audiometric progression, and to map DFNA75-specific vs DEDDFA-specific variant domains.

2. **Allele-specific functional assays:** Introduce p.Arg171Cys and other DFNA75 alleles into cochlear-relevant cell/organoid systems (inner-ear organoids, iPSC-derived otic cells) and assess TRRAP scaffold integrity, HAT-complex assembly, and target-gene acetylation/ expression.
3. **Knock-in mouse model:** Generate a *Trrap* p.Arg171Cys knock-in (or cochlea-conditional) mouse to test whether it recapitulates progressive SNHL and to localize affected cochlear cell types (hair cells vs supporting cells).
4. **Structure–function mapping:** Use AlphaFold/cryo-EM of TRRAP within STAGA/TIP60 to model how DFNA75 versus DEDDFA missense positions perturb specific protein interfaces.
5. **Cochlear cell-type resolution:** Apply single-cell/spatial transcriptomics of the developing inner ear (mouse/zebrafish) under TRRAP perturbation to define the cell populations and gene programs disrupted.
6. **Standardize diagnostics:** Ensure TRRAP is included on comprehensive ADNSHL gene panels and provide ACMG-based reclassification support for new variants; implement cascade testing and genetic counseling protocols for identified families.
7. **Clinical outcome tracking:** Follow hearing-aid and cochlear-implant outcomes in genetically confirmed DFNA75 patients to establish genotype-informed rehabilitation expectations.

Report compiled from a 5-iteration autonomous investigation: 7 confirmed findings, 20 papers reviewed. Evidence types are indicated throughout (human clinical, model organism, in vitro, computational/inferred).
